21BM 407 BIOLOGY FOR ELECTRONICS ENGINEERS

UNIT III	NERVOUS AND URINARY SYSTEM
Nervous System- Neuron anatomy, Types, Central Nervous system, peripheral nervous system, function of brain-four lobs, Urinary System –Anatomy and function of kidney, urine formation- 4 steps.	

NERVOUS SYSTEM

The brain is like a computer that controls the body's functions, and the nervous system is like a network that relays messages to parts of the body. The nervous system is a human body's command centre. Originating from the brain. The nervous system guides almost everything you do, think, say or feel. it controls body's movements, thoughts and automatically responses to the world arounds you. It also controls the body systems and processes, such as digestion, breathing and sexual development (puberty). Diseases, accidents, toxins and the natural aging process can damage your nervous system.

The nervous system is responsible for every aspect of your life, including

- Thoughts, memory, learning, and feelings.
- Movements, such as balance and coordination.
- Senses- your brain interprets what you see, hear, taste, touch and feel.
- Sleep, healing and aging.
- Heartbeat and breathing patterns.
- Response to stressful situations.
- Digestion, as well as how hungry and thirsty you feel.
- Body processes, such as puberty.

A vast network of nerves sends electrical signals to and from other cells, glands, and muscles all over the body. These nerves receive information from the world around you. Then the nerves interpret the information and control your response. It's almost liked an enormous information highway running throughout your body.

The nervous system is classified in to three

- Sensory nervous system
- Integrative nervous system
- Motor nervous system

Sensory nervous system

Millions of sensory receptors detect changes, called stimuli, which occur inside and outside the body. They monitor such things as temperature, light, and sound from the external environment. Inside the body, the internal environment, receptors detect variations in pressure, pH, carbon dioxide concentration, and the levels of various electrolytes. All of this gathered information is called sensory input.

Integrative nervous system

Sensory input is converted into electrical signals called nerve impulses that are transmitted to the brain. There the signals are brought together to create sensations, to produce thoughts, or to add to memory; Decisions are made each moment based on the sensory input. This is integration.

Motor nervous system

Based on the sensory input and integration, the nervous system responds by sending signals to muscles, causing them to contract, or to glands, causing them to produce secretions. Muscles and glands are called effectors because they cause an effect in response to directions from the nervous system. This is the motor output or motor function.

NEURON ANATOMY

Neurons are also known as nerve cells, send and receive signals from the brain. In a human body around 86 billion neurons or nerve cells are available. However, their structure and functionality are unique. The creation of new nerve cells is called neurogenesis. The activity of neurons has been started when you were as an embryo.

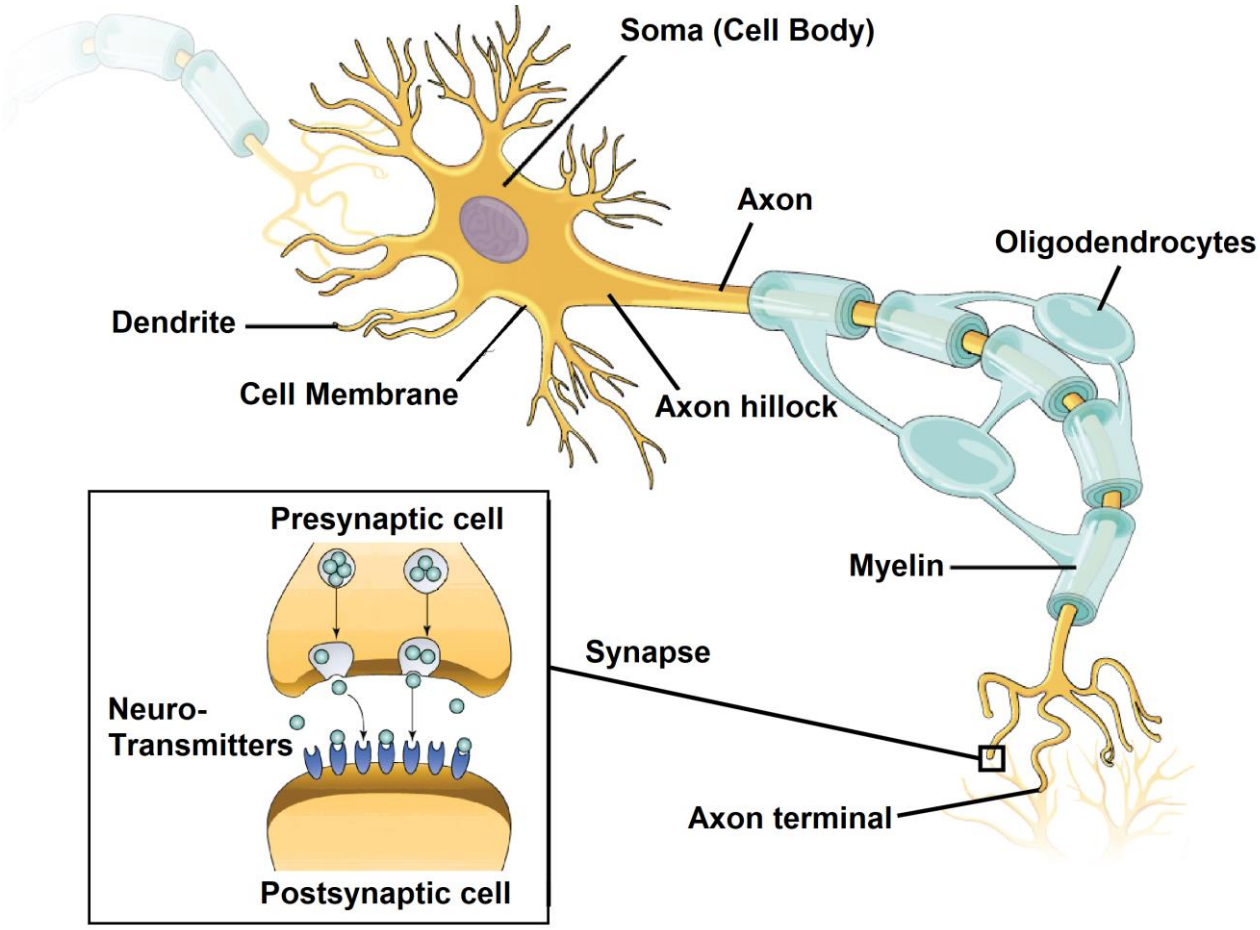

Fig. 3.1 Structure of a neuron

The Structure of a neuron is shown in the fig 3.1 Neurons vary in shape, size, and structure depending on their role and location. However, nearly all neurons have three essential parts: a cell body, an axon, and dendrites.

Cell body

Cell body also known as soma, the cell body is the core section of the neuron. The cell body contains genetic information, maintains the neuron's structure, and provides energy to drive activities. The neurons cell body (soma) contains a nucleus and specialized organelles. It's enclosed by a cell membrane it does two functions ie. It protects the neuron and it interacts with its immediate near by surroundings.

Axon

Each neuron usually have one main axon. An axon is a long, tail-like structure. It joins the cell body at a specialized junction called the axon hillock. Many axons are insulated with a fatty substance called myelin. Myelin helps axons to conduct an electrical signal. oligodendrocytes assemble myelin. Towards the end of axon, the axon splits up into many branches known as axon terminals (or nerve terminals). These axon terminals make connections on target cells.

Dendrites

Dendrites are fibrous roots that branch out from the cell body. Like antennae, dendrites receive and process signals from the axons of other neurons. Neurons can have more than one set of dendrites, known as dendritic trees.

Number of dendrites present in each neuron is depends on their role. For instance, Purkinje cells are a special type of neuron found in a part of the brain called the cerebellum. These cells have highly developed dendritic trees which allow them to receive thousands of signals.

Synapses

Neuron-to-neuron connections are made onto the dendrites and cell bodies of other neurons. These connections, known as synapses, from brain an information is carried from the first neuron called presynaptic neuron, to the target neuron called the postsynaptic neuron. The information is transmitted in the form of chemical messengers called neurotransmitters. From the presynaptic cell, Neurotransmitter molecules cross each synapse and reaches the membrane receptors on the postsynaptic cell, conveying an excitatory or inhibitory signal.

The synaptic connections between neurons and skeletal muscle cells are generally called neuromuscular junctions, and the connections between neurons and smooth muscle cells or glands are known as neuroeffector junctions.

Types of neurons

Neurons vary in structure, function, and genetic makeup. There are thousands of different types of neurons, However, there are five major neuron forms found in the various living organisms.

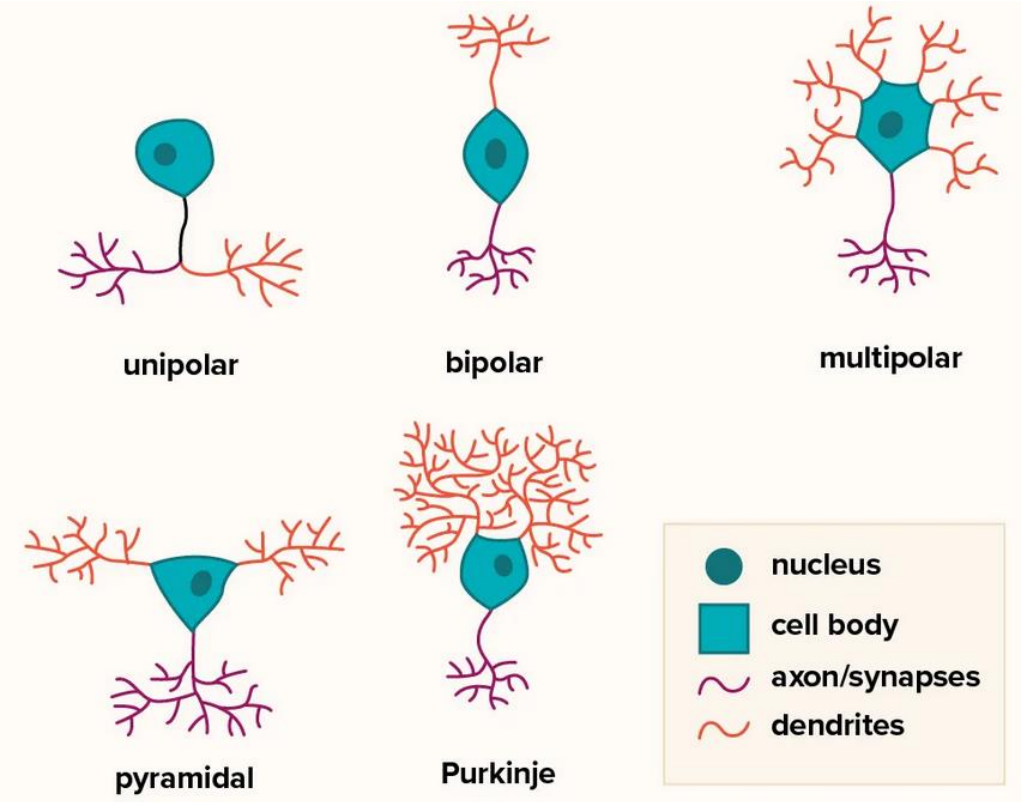

Fig.3.2 Types of neurons

Unipolar neurons - have a single axon. Usually found in invertebrate species,

Bipolar neurons. Bipolar neurons have two extensions extending from the cell body. At the end of one side is the axon, and at the other side is dendrites. These types of neurons are mostly found in the retina of the eye. But they can also be found in parts of the nervous system that help the nose and ear function.

Multipolar neurons. These neurons have a single axon and symmetrical dendrites that extend from it. This is the most common form of neuron in the central nervous system.

Pyramidal neurons. These neurons have one axon but several dendrites to form a pyramid type shape. These are the largest neuron cells and are mostly found in the cortex. The cortex is the part of the brain responsible for conscious thoughts.

Purkinje neurons. Purkinje neurons have multiple dendrites that fan out from the cell body. These neurons are inhibitory neurons, meaning they release neurotransmitters that keep other neurons from firing.

TYPES OF HUMAN NERVOUS SYSTEM

The human nervous system is broably classified in to two types

- Central Nervous System (CNS)
- Periperal Nervous System (PNS)

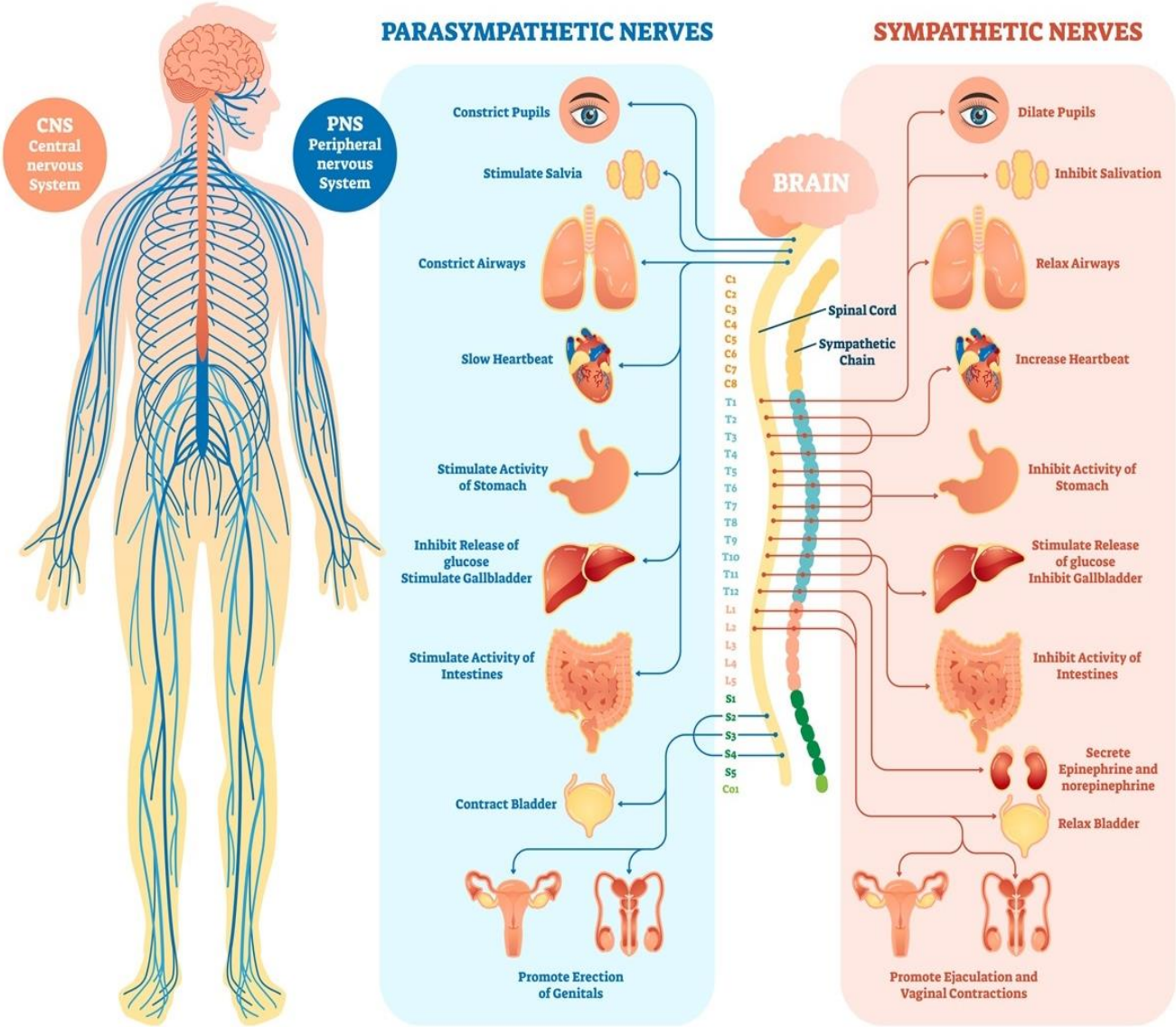

Fig.3.3 Human Nervous System

The central nervous system

The central nervous system consists of the brain and spinal cord. It is referred to as “central” because it combines information from the entire body and coordinates activity across the whole organism. The retina, optic nerve, olfactory nerves (sensory nerve and conveys the sense of smell), and olfactory epithelium (membranous tissue located inside the nasal cavity) are considered to be part of the CNS alongside the brain and spinal cord. This is because they connect directly with brain tissue without intermediate nerve fibers.

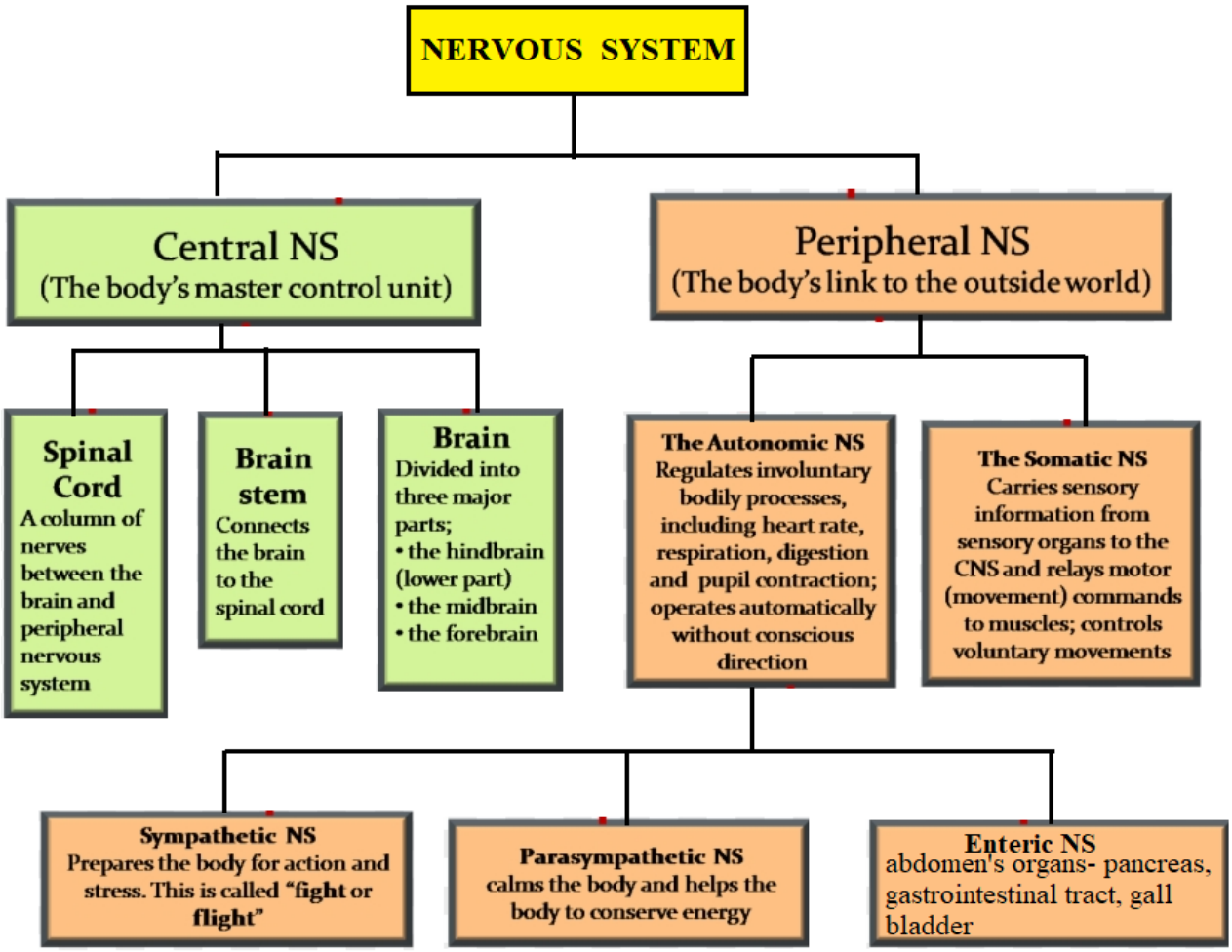

Fig.3.4 Human Nervous System classification

Brain

The brain is the most complex organ in the body and uses 20 percent of the total oxygen we breathe in. The brain consists of an estimated 100 billion neurons, with each connected to thousands more.The brain can be divided into four main lobes: temporal lobe, parietal lobe, occipital lobe and frontal lobe.

- **Temporal lobe**: For processing sensory input, long term memory, language perception and assigning it emotional meaning.
- **Occipital lobe** : visual processing region of the brain, housing the visual cortex.
- **Parietal lobe:** integrates sensory information from all 5 sense organs including touch, spatial awareness, and navigation.
- **Frontal lobe :** positioned at the front of the brain, it is involved in attention, reward, short-term memory, motivation, and planning.

Spinal cord

The spinal cord, running almost the full length of the back, carries information between the brain and body, but also carries out other tasks.From the brainstem, where the spinal cord meets the brain, 31 spinal nerves enter the cord. Along its length, it connects with the nerves of the peripheral nervous system (PNS) that run in from the skin, muscles, and joints. Motor commands from the brain travel from the spine to the muscles and

sensory information travels from the sensory tissues — such as the skin — toward the spinal cord and finally up to the brain.

The spinal cord contains circuits that control certain reflexive responses, such as the involuntary movement your arm might make if your finger was to touch a flame.The circuits within the spine can also generate more complex movements such as walking. Even without input from the brain, the spinal nerves can coordinate all of the muscles necessary to walk.

The peripheral nervous system

The Peripheral nervous system is made up of two parts:

- Somatic nervous system
- Autonomic nervous system

Somatic nervous system

The somatic nervous system controls volantary muscles. Itconsists of peripheral nerve fibers that pick up sensory information or sensations from the peripheral organs (those away from the brain like limbs) and carry them to the central nervous system.

These also consist of motor nerve fibers that come out of the brain and take the messages for movement and necessary action to the skeletal muscles. For example, on touching a hot object, the sensory nerves carry information about the heat to the brain, which in turn, via the motor nerves, tells the muscles of the hand to withdraw it immediately.

The whole process takes less than a second to happen. The neuron's cell body that carries the information often lies within the brain or spinal cord and projects directly to a skeletal muscle.

Autonomic Nervous System

Another part of the nervous system is the Autonomic Nervous System. It has three parts:

- The sympathetic nervous system
- The parasympathetic nervous system
- The enteric nervous system

The sympathetic nervous system controls the nerves of the body's inner organs on which humans have no conscious control. This includes the heartbeat, digestion, breathing (except conscious breathing), etc.

The parasympathetic nervous system controls the smooth involuntary muscles of the (internal organs) and glands and cause them to function and secrete their enzymes.

The enteric nervous system is the third part of the autonomic nervous system. It is a complex network of nerve fibers that innervate the abdomen's organs like the gastrointestinal tract, pancreas, gall bladder, etc. It contains nearly 100 million nerves.

ANATOMY OF HUMAN BRAIN

The brain is the most complex organ in the human body, contains 15–33 billion neurons, each of which is connected to thousands of other neurons. In total, around 100 billion neurons and 1,000 billion glial (support) cells make up the human brain. Our brain uses around 20 percent of our body's total energy.

Brain Cells

There are two types of cells in your brain, neurons and glial cells (glia - Greek word for glue). For a long time, biologists have thought that the neurons were the only cells that controlled our bodies and were also where our memories are kept. Glial cells present in the brain to support neurons, to insulate neurons, provides nutrition to neuron, and do basic housekeeping for neurons.

Support: Glia cells act as a physical support and protection for neurons. They also help keep the blood-brain barrier which prevents toxic chemicals in the blood from entering the brain.
Insulation: Glia cells can create myelin, a fatty substance that helps insulate the axons of neurons. This helps keep electrical signals inside the neuron and helps them move faster.
Nutrition: Glia cells help keep the environment around neurons in balance and make sure the right nutrients are available for neurons.
Housekeeping: Glia cells can prevent the buildup of toxic chemicals, help destroy viruses and bacteria, and get rid of dead neurons

Four lobes of brain

The brain is the central control module of the body and coordinates activity. From physical motion to the secretion of hormones, the creation of memories, and the sensation of emotion.To carry out these functions, some sections of the brain have dedicated roles. However, many higher functions — reasoning, problem-solving, creativity — involve different areas working together in networks.The brain is roughly split into four lobes:

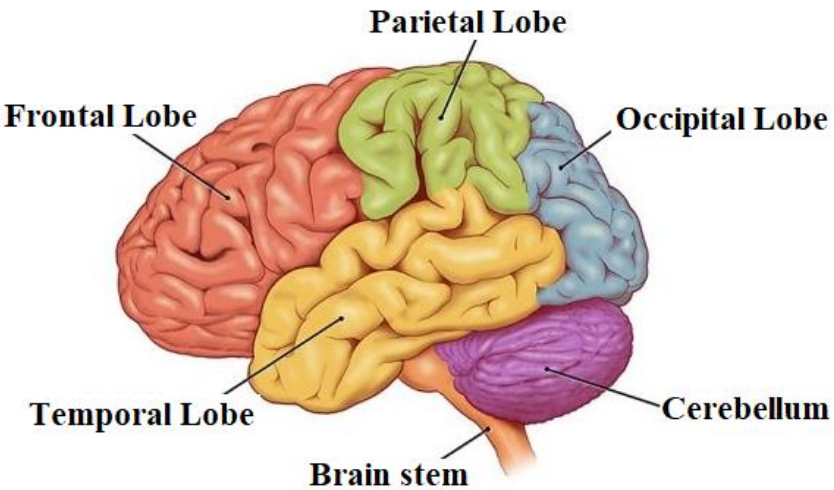

Fig.3.5 four lobes of brain

Temporal lobe: For processing sensory input, long term memory, language perception and assigning it emotional meaning.
Occipital lobe : visual processing region of the brain, housing the visual cortex.
Parietal lobe: integrates sensory information from all 5 sense organs including touch, spatial awareness, and navigation.
Frontal lobe : positioned at the front of the brain, it is involved in attention, reward, short-term memory, motivation, and planning.

Regions and function of human brain

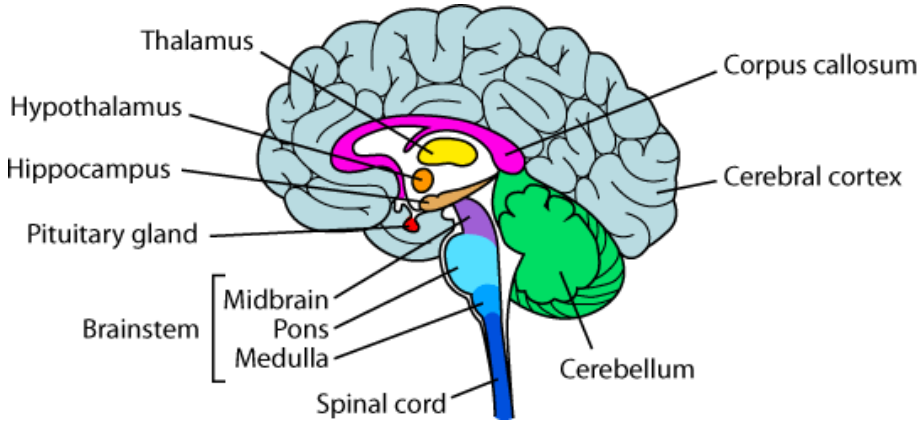

Fig.3.6 Regions and function of the human brain

Cerebellum: Main job- For proper movement and present At the back of the brain.

The cerebellum helps you make smooth, coordinated movements. Walking, running, writing . . . without the cerebellum, movement would not happen properly.

Cerebral cortex: Main job-sensory integration, thought, voluntary movement, language, reasoning, perception.

The outermost largest part of the brain. It helps for thinking, reading, mathematical calculations, this part of the brain also controls voluntary muscles. These are the muscles used to click a computer mouse button, ride a bike, run a race, or kick a ball.
the right and left sides of the cerebral cortex specialize in different work. The right side works on abstract things like colours, shapes, and music. The left side is busy with math and speaking and is the analytical part. It is also known that the right side of your brain controls the left side of the body and the left side of the brain controls the right.

Corpus callosum: Main jobs- connecting the left and right brain

Connecting the right side of the brain with the left side is the corpus callosum. This lets the two sides of the brain talk to each other.

Thalamus: Main job: sensory integration, motor integration, sleep, consciousness.

It is positioned in the centre of the brain; the thalamus receives sensory and motor input and relays it to the rest of the cerebral cortex. It is involved in the regulation of consciousness, sleep, awareness, and alertness.

Hypothalamus: Main job: body temperature, emotions, hunger, thirst,

It is present just above the brain stem and it is roughly the size of an almond, The hypothalamus is the thermostat for your body. When you get hot it sends signals to your skin to sweat. If you get cold it makes you shiver and get goose bumps. the hypothalamus secretes a number of neurohormones and influences body temperature control, thirst, and hunger.

Hippocampus: Main job - creating memories

Hippocampus is located deep within your brain; the hippocampus works with the cerebral cortex to create memories. From five minutes ago to five years ago, the hippocampus is responsible for helping you remember everything that happens in your life. The hippocampus also helps you remember how to navigate, or move around, your environment based on how you have moved around in the past.

Pituitary gland: Main job: controls hormones

Pituitary gland is a tiny part of the brain is about the size of a pea. It is smallest part of the brain. It releases special hormones that help you grow and also the one that controls when and how fast we grow. It also is at work during puberty. The pituitary gland also uses hormones to control how much sugar and water is in your body. It also is one of the areas that controls the body's metabolism. It helps control the digestion of food, breathing, and moving your blood around. Without strict control of your body's metabolism, it would not be possible to stay alive.

Brainstem: Main jobs- breathing, heart rate, blood pressure

The brainstem is located below the cerebellum and connects the brain to the spinal cord. It has the job of running all the involuntary muscles. These are the muscles you don't consciously control, such as the muscles which make your heart beat. The brainstem is a collection of three areas of the brain. Namely **Medulla, Pons**, and the **Midbrain**.

- **Medulla oblongata:** extending below the skull, it is involved in involuntary functions, such as vomiting, breathing, sneezing, and maintaining the correct blood pressure.
- **Pons:** Contains some relays, responsible for facial expressions and auditory system.
- **Midbrain** is the topmost part of the brainstem, the connection central between the brain and the spinal cord. The midbrain serves important functions in motor movement, particularly eye movements and in auditory and visual processing.

Together, these three parts of the brain help keep us alive by controlling our breathing, digestion, and blood circulation.

spinal cord: Main job: moving messages from the brain to the rest of the body.

Spinal cord is the main information highway. All the information from parts of the body moves from and to the brain through the spinal cord. It is connected to the brain through the brainstem. It is protected by the vertebral bones that make up the backbone.

URINARY SYSTEM

The body takes nutrients from food and converts them to energy. After the body has taken the food components that it needs, waste products are left behind in the bowel and in the blood. The main function of the urinary system is to filter blood and create urine as a waste by-product. The urinary system, also known as the urinary tract or renal system.

The kidney and urinary systems help the body to eliminate liquid waste called urea, and to keep chemicals, such as potassium, sodium, and water in balance condition. Urea is produced when foods containing protein, such as animal meat, and certain vegetables, are broken down in the body. Urea is carried in the bloodstream to the kidneys, where it is removed along with water and other wastes in the form of urine. The urinary system works with the lungs, skin and intestines to maintain the balance of chemicals and water in the body. Adults eliminate about 27 to 68 ounces of urine (800 to 2,000 milli liters) per day based on typical daily fluid intake of 68 ounces (2 liters),

Urinary system parts and their function

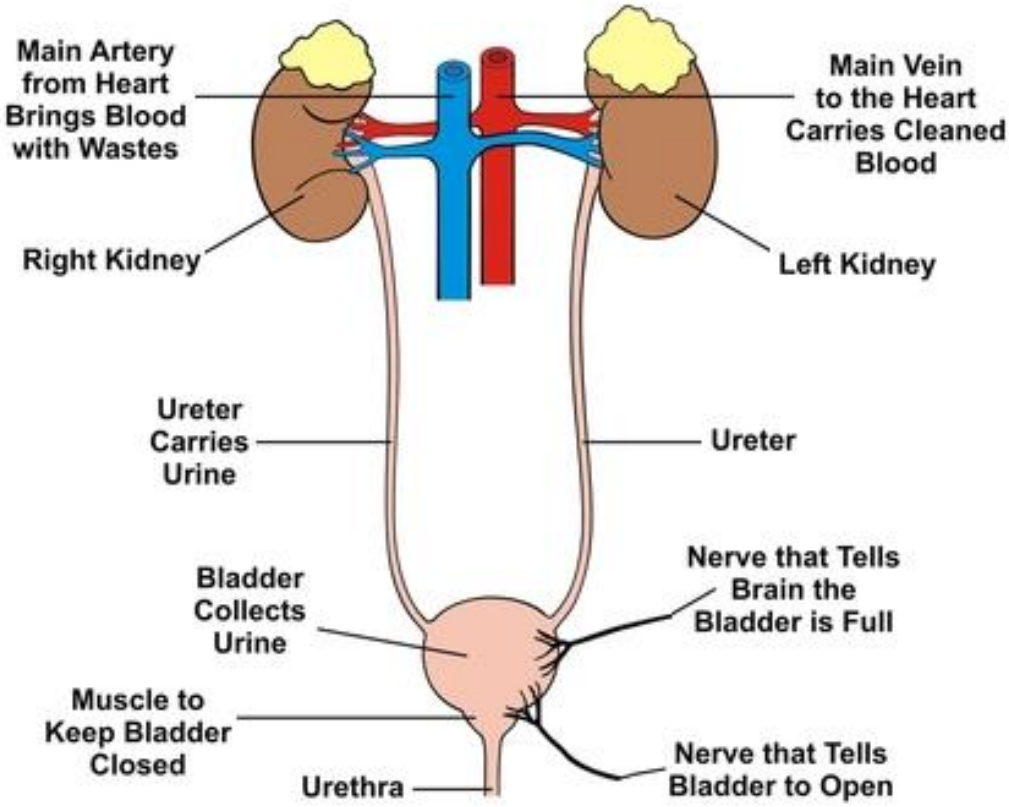

Fig.3.7. Parts of a urinary System

The urinary system is divided into two parts. The upper urinary system and the lower urinary system. The upper urinary system includes the kidneys and ureters. The lower urinary system includes the bladder and urethra.

Kidney:

A pair of purplish-brown kidneys are bean shaped located below the ribs toward the middle of the back. Every minute, approximately 1300 mL of blood enter the kidneys, 1299 mL leave the kidney, and approximately 1 mL leaves the body as urine. no matter how much we eat or drink. If the body is dehydrated, the kidneys put less water in the urine. When the blood becomes too acidic, the kidneys remove more acid from the blood and excrete as urine. Also, the kidney serves many functions.

The main function of the kidney is to:

- Remove waste products and drugs from the body.
- Balance the body's fluids.
- Release hormones to regulate blood pressure.
- Control production of red blood cells.
- Regulate the acid-base balance.

Ureters:

Ureters are narrow tubes carry urine from the kidneys to the bladder. The ureters are about 8 to 10 inches long (20 to 25 cm) Muscles in the ureter walls continually tighten and relax forcing urine downward, away from the kidneys. If urine backs up, or is allowed to stand still, a kidney infection can develop. Small amounts of urine are emptied into the bladder from the ureters about every 10 to 15 seconds.

Bladder:

Bladder is a triangle-shaped, hollow organ is located in the lower abdomen. It is held in place by ligaments that are attached to other organs and the pelvic bones. The bladder's walls relax and expand to store urine, and contract and flatten to empty urine through the urethra. The typical healthy adult bladder can store up to two cups of urine for two to five hours.

Urethra:

Urethra is a tube-like structure present at the end of the bladder. It allows urine to pass outside the body. The brain signal makes the bladder muscles to tighten, which squeezes urine out of the bladder. At the same time, also the brain signal makes the bladder muscles to relax to let urine exit the bladder through the urethra. When all the signals occur in the correct order, normal urination occurs.

ANATOMY OF KIDNEY

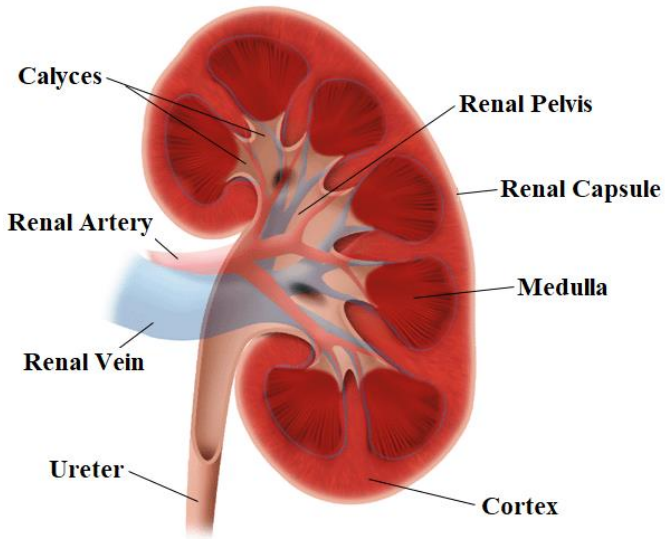

Fig.3.8. Anatomy of Kidney

Renal Capsule: An outer membrane that surrounds the kidney; it is thin but tough and fibrous helps to protect the kidney.

Cortex: The outer region of the kidney; extensions of the cortical tissue, contains about one million blood filtering nephrons.

Nephron: Nephrons are the filtration units in the kidneys.

Medulla: The inner part of the kidney contains 8-12 renal medullary pyramids. It acts as a urine collecting ducts.

Calyces: The extension of the renal pelvis; they channel urine from the pyramids to the renal pelvis.

Renal Pelvis: Basin-like area that collects urine from the nephrons (the kidney's filtration system), it narrows into the upper end of the ureter.

Ureters: Ureters are narrow tubes carry urine from the kidneys to the bladder. The ureters are about 8 to 10 inches long (20 to 25 cm) Muscles in the ureter walls continually tighten and relax forcing urine downward, away from the kidneys. If urine backs up, or is allowed to stand still, a kidney infection can develop. Small amounts of urine are emptied into the bladder from the ureters about every 10 to 15 seconds.

Renal Arteries: A pair of arteries that branch off of the aorta bringing waste-filled blood into the kidney for filtering. Each minute, the kidneys receive 20% of the blood pumped by the heart.

Renal Veins: Remove the filtered blood from the kidneys to the inferior vena cava.

FUNCTION OF A KIDNEY AND URINE FORMATION

As mentioned, prior, the kidneys remove waste products from the body (creatinine, urea, ammonia, etc.) while ensuring that essential substances are retained.

Each of your kidneys is made up of about a million filtering units called nephrons. Each nephron consists of glomerulus, and tubule. The glomerulus filters your blood, and the tubule does two main functions 1. Reabsorb the needed substances to your blood and 2. Secretion i.e., removal of wastes.

The kidneys collect and get rid of waste from the body in 3 steps:

- Glomerular filtration
- Tubular reabsorption
- Tubular secretion

Glomerular filtration – Filtrate is made as the blood is filtered through a collection of capillaries in the nephron called glomeruli.

The glomerular filtration rate (GFR) is the rate at which the glomeruli filter the blood. The normal GFR is 120 ml/minute. The most accurate measure of the GFR is done by measuring creatinine clearance. Clearance is the complete removal of a substance from the blood. Creatinine clearance is a good measure of filtration rate because creatinine (a waste product of the body) is filtered from the blood but is not reabsorbed by the tubules.

Tubular reabsorption – The tubules in the nephrons reabsorb the filtered blood in nearby blood vessels.

Tubular secretion – The remaining filtrate which contains waste product passes through the tubules to the collecting ducts and is then taken to the bladder via the ureters.

NEPHRON ANATOMY

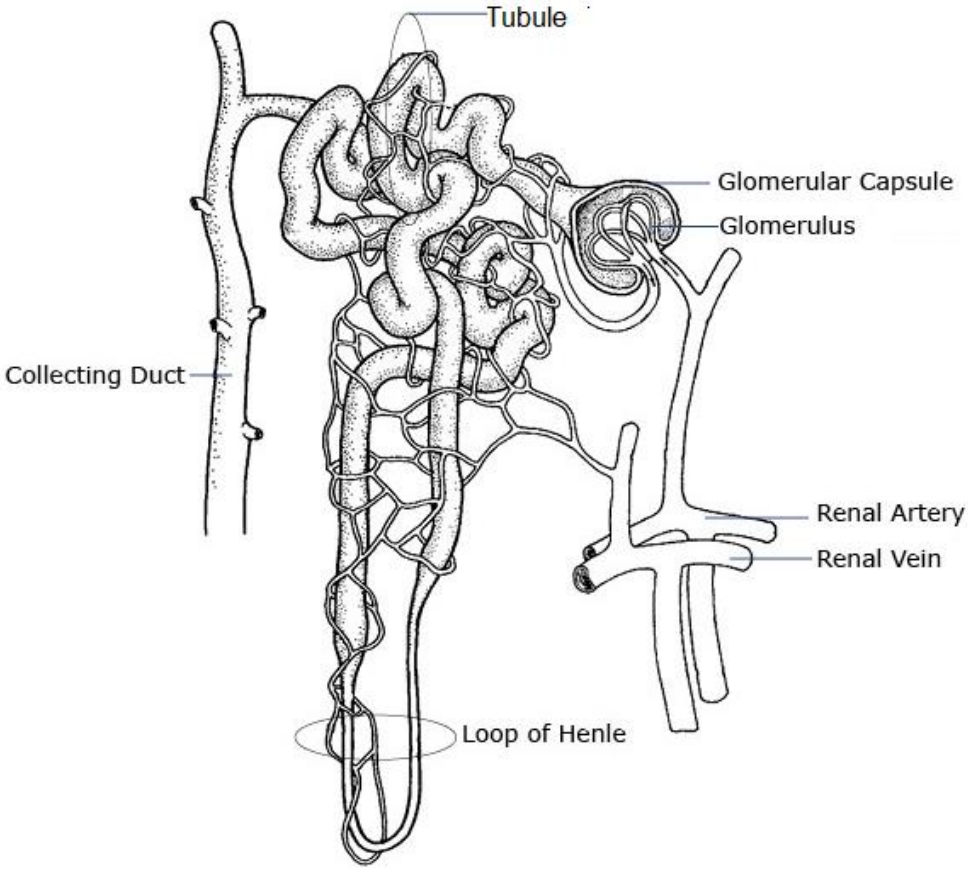

Fig.3.9 Nephron Anatomy

The nephron is the filtration unit of the kidney. It is the basic structural and functional unit of the kidney and cannot be seen by the naked eye. Each kidney has about 1 million nephrons. The walls of the nephron are made of a single layer of epithelial cells. Blood containing urea and metabolic waste products enters the kidney from the bloodstream. The blood is filtered by the nephrons which is present in the kidney to remove wastes while leaving behind the blood cells and vital proteins and chemicals. The kidney also functions to help maintain body fluid and electrolyte homeostasis (balance).

- **Renal Artery** – brings waste-filled blood from the aorta to the kidney for filtering.

- **Renal Vein** – when filtration is complete, blood leaves the nephron to join the renal vein, which removes the filtered blood from the kidney and delivers it back to the inferior vena cava.

- **Glomerulus** – each glomerulus is a cluster of blood capillaries surrounded by a Bowman's capsule. The filtration process is done in the glomerulus.

- **Tubule** – It is very nearest to glomerulus; have permeable cell membranes that reabsorb vital metabolites and electrolytes into nearby capillaries. The remaining filtrate which contains waste product which is passed through the tubules to the collecting ducts.

- **Collecting Duct** – collects the filtrate, and is then taken to the bladder via the ureter.

- **Loop of Henle** – It is a long U-shaped portion of the tubule that conducts urine within each nephron of the kidney. The main function of the loop of Henle is to recover water and sodium chloride from urine.

What is urine made of?

Urine is made of water, urea, electrolytes, and other waste products. The exact contents of urine vary depending on how much fluid and salt you take in, your environment, and your health. Some medicines and drugs are also excreted in urine and can be found in the urine.

- 94% water
- 5% urea
- 1% sodium
- 0.5% chloride
- 0.25% potassium
- 0.25% phosphate
- 0.25% sulphate
- 0.15% creatinine
- 0.1% uric acid

As mentioned, urine is formed in the nephrons by a three-step process: glomerular filtration, tubular re-absorption, and tubular secretion. The amount of urine varies based on fluid intake and one's environment.